

SELF-HEALING ELECTRONICS: On The Way



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As electronic devices are evolving to perform more sophisticated tasks, manufacturers are packing as much density onto a chip as possible. However, such density causes reliability problems such as failure stemming from fluctuating temperature cycles as the device operates or fatigues. A failure at any point in the circuit can shut down the whole device.

However, there is only so much you can do to manually repair a modern sophisticated electronic device, as sometimes you cannot even get to the insides of such a device. For example, in a multilayered integrated circuit, there is no way to open it up. Normally, you would just replace the whole chip.

This is true for a battery, too. You cannot pull a battery apart to try to find the source of the failure. Most consumer devices are meant to be replaced with some frequency, adding to electronic waste issues, but in many important applications such as instruments or vehicles for space or military functions, electrical failures cannot be replaced or repaired.

Breaking of materials with minor or major faults in electronic devices leads to their malfunctioning. These material faults can be rectified by an expert technician on the job, but this would lead to a vicious circle of exploitation and might eventually force you to discard the faulty electronic device prematurely. When one tiny circuit within

an integrated chip cracks or fails, the whole chip or the whole device is a loss. But what if it could fix itself, and fix itself so fast that you never even know there was a problem?

Electronic materials have been a major stumbling block in the advancement of flexible electronics because existing materials do not function well after breaking and re-making. Electronic devices are subjected to mechanical deformation over time, which could destroy or break these.

Self-healing materials are a class of smart materials that have the structurally-incorporated ability to repair the damage caused by mechanical usage over time. The inspiration comes from biological systems, which have the ability to heal after being wounded. A team of engineers has developed a self-healing system that restores electrical conductivity to a cracked circuit in less time than it takes to blink. Rather than having to build in redundancies or a sensory diagnostics system, this material is designed to take care of the problem itself.

Electronics requirements from self-healing materials

Self-healable materials are those that, after withstanding physical deformation such as being cut in half, naturally repair themselves with little to no external influence. The new electronic material created, however, can heal all its functions automatically even after

Researchers have developed a flexible electronic material that self-heals to restore many functions, even after multiple breaks. Here, the material is shown being cut in half. The healed material is still able to stretch and hold weight (Image courtesy: Qing Wang/Penn State)

